

Rosh Ho

r.ho@columbia.edu | +1 929-900-4405

@roshho | Rosh Ho | roshho.com

Software Engineer | Backend & Machine Learning Systems

Education

Columbia University

Sep 2021 - Aug 2025

Computer Science

New York, New York

- **Paper:** ReGlove [arXiv:2512.11824](#) (IEEE BioRob 2026); Joint Inverse & Forward Dynamics [arXiv:2503.12297](#)
- **Awards:** Columbia DevFest Hackathon 2nd; Teacher's College College Mental Health x AI Hackathon 1st; UN Millennium Fellowship
- Coursework: Deep Learning with Computer Vision, Natural Language Processing, Computational Aspects of Robotics, Robotics Manufacturing, Embedded Systems, Databases, Operating Systems

Experience

Clinra RCM

Jul 2025 - Current

AI/ML Software Engineer

New York

- Built a **two-stage Medicare claim denial prediction** pipeline using **XGBoost** and a deterministic rules engine over CMS claims data, achieving **74% PR-AUC** and **62% F1** across a 12-provider pilot reducing avoidable denials by **18%**
- Engineered a **RAG-based appeal generation** system using **Amazon Nova** over CMS LCD/NCD policy documents, with **prompt guardrails** and post-generation scrubbing, reducing appeal drafting time by **40%** across a 3-hospital pilot
- Contributed to production **React + Next.js** frontend for Medicare revenue cycle platform, implementing role-based dashboard rendering for physician and RCM analyst personas with real-time denial risk visualizations via **Recharts** and **D3.js**
- Headed startup to over \$65k pre-seed funding and won grant from New York State Energy Research and Development.

Robotic Manipulation and Mobility Lab at Columbia University

Feb 2024 - May 2025

Robotics Research Assistant

Columbia University

- Developed **robotics control modules** in **C++** and **Python** within **ROS/Linux** environment to translate human grasping patterns into **real-time autonomous manipulation routines**.
- Built modular **data-collection systems** that reduced preprocessing time by **40%**, streamlining the mapping of human grasping patterns to robotic actuation.

LINE

May 2024 - Sep 2024

Backend Software Engineering Intern

Taipei, Taiwan (ROC)

- Engineered **high-performance backend** for dynamic ad placement on streaming platforms using **Go**, **AWS**, and **MySQL**, **reducing request latency by 23%** in a **latency-sensitive production environment**.
- Designed and implemented **automated unit-testing pipeline** in **Go**, improving **API reliability** and **scalability** for a system serving **millions of daily requests**.

Blackstone

May 2023 - Aug 2023

Software Engineering Intern

New York, New York

- Built a **Python-based monitoring pipeline** for simulated data center systems, applying **AI-driven anomaly detection** to sensor data (temperature, power, throughput).
- Deployed results to a **Grafana dashboard** with **Prometheus** and automated weekly reporting using **Airflow**, improving operational insights and reducing manual analysis.

Singapore Armed Forces

Sep 2018 - Sep 2020

Automation Engineer (Corporal)

Singapore, Singapore

- Developed Python and Bash scripts to automate ETL pipelines over **SAP** data exports, replacing manual workflows across **8,000+** medical records with validated loads into a **MySQL** database.
- Built **UiPath RPA bots** with custom error-handling, retry logic, and **SQL stored procedures** to standardize reporting and reduce manual analysis time by **60%**.

Skills

Programming Languages Python, C/C++, Go, SQL

Technologies Linux, Pytorch, Langchain, Git, ROS, MySQL, AWS, Jupyter, Amazon Bedrock, CAD, SCRUM

Specialties Low-latency systems, algorithm optimization, automated testing, data analysis system design, problem solving, debugging